

Tasks 3, 5: see the repository

Repository: <https://github.com/JangwonLim/bmi500>

Task 3 (vibe coding): `matvec_multiply.py`, `matvec_test.py`, `AI_code_gen.txt`

Task 5 (code review): `manual_code_review.txt`, `ai_code_review.txt`

Key Performance Findings

KEY PERFORMANCE FINDINGS

Coarse-grain usage (wall / user CPU / system CPU / maximum memory):

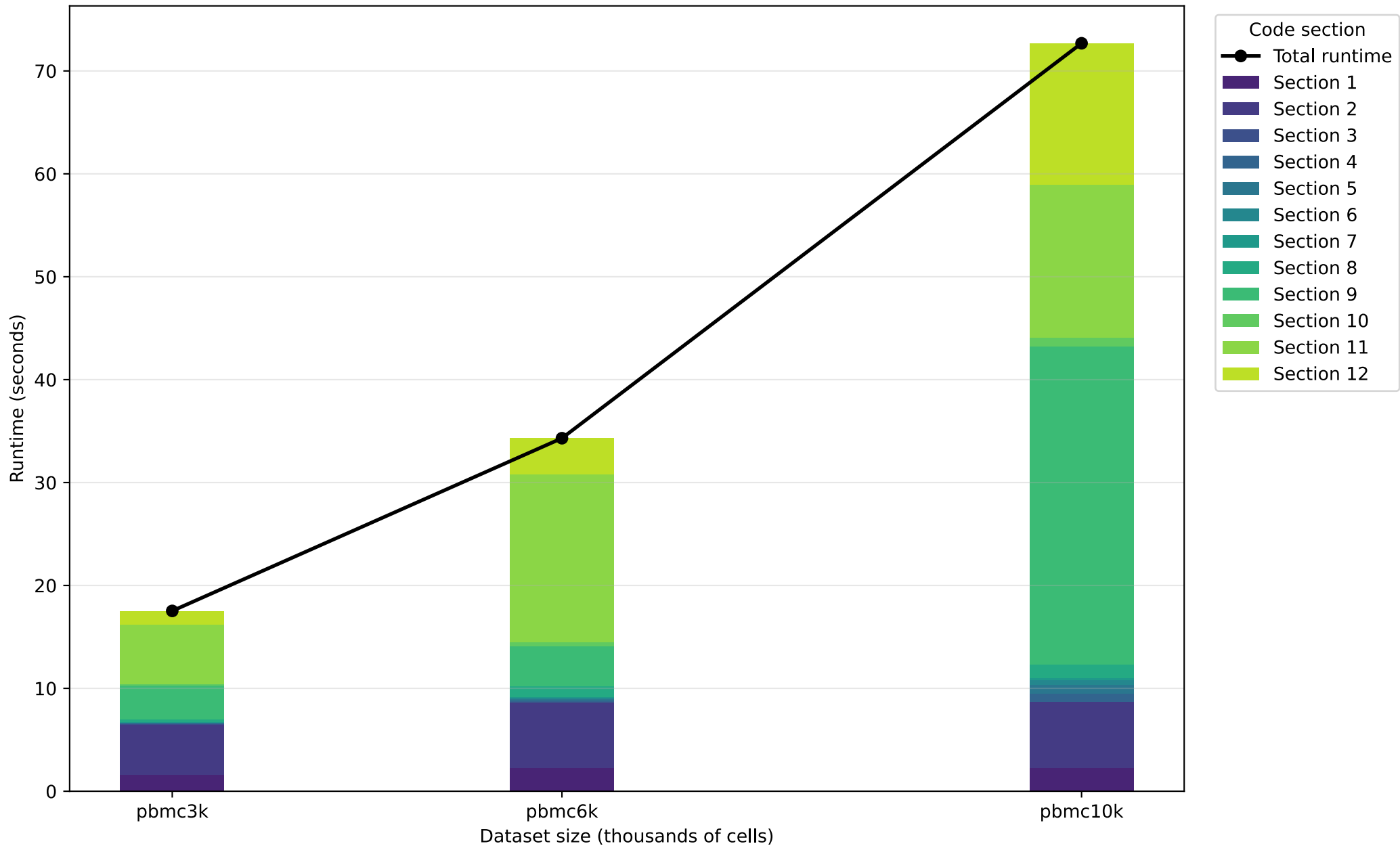
pbmc3k: 0:19.52 / 18.76 s / 1.60 s / 868 MB

pbmc6k: 0:26.62 / 34.84 s / 2.49 s / 962 MB

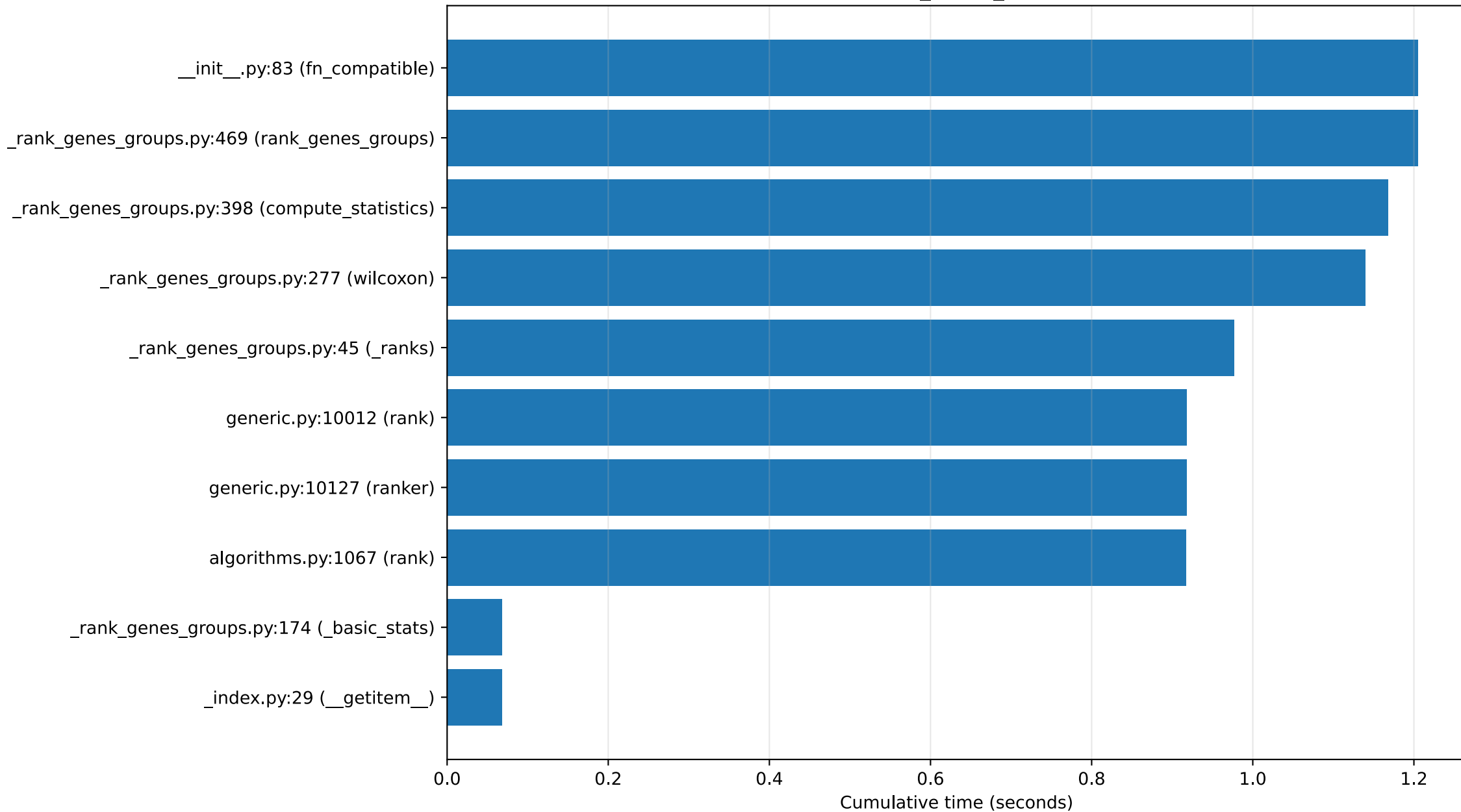
pbmc10k: 0:56.27 / 63.20 s / 6.63 s / 1846 MB

I believe the major bottleneck happens at the section 9 (neighborhood graph) with a bigger dataset size as shown in the stacked bar chart. Runtime grew significantly from the smaller datasets, so if the data size were to grow to 20K, the expected running times would be approximately 2.5 times. Since the entire runtime for 10K dataset is around 60-70 seconds, the expected runtime would be around 150-175 seconds. If it's not linear, which is most likely, then it might take much longer.

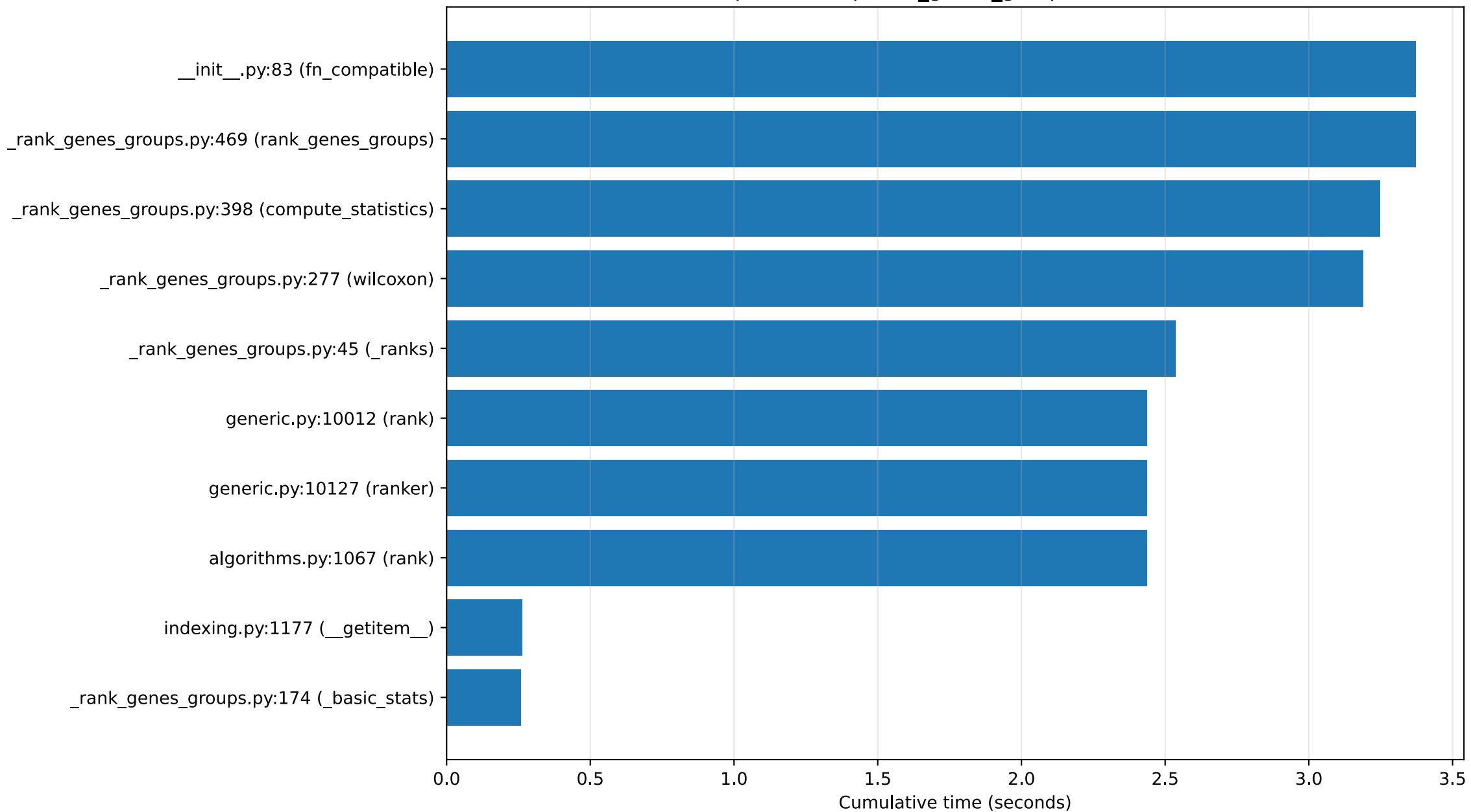
Instrumented runtime growth by dataset



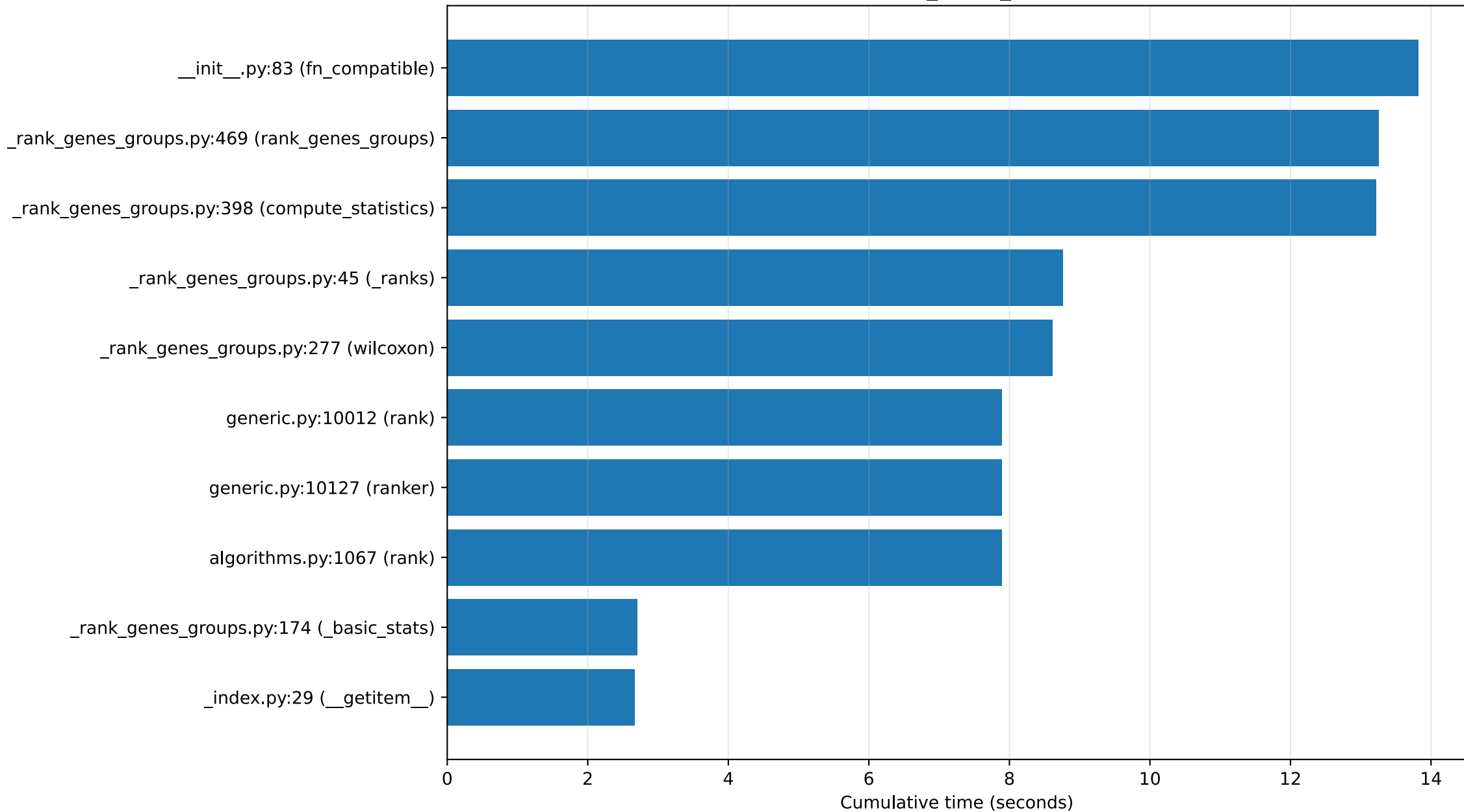
pbmc3k: top rank_genes_groups functions



pbmc6k: top_rank_genes_groups functions



pbmc10k: top rank_genes_groups functions



pbmc3k: fine-grain profiling output

Dataset: pbmc3k

cProfile of sc.tl.rank_genes_groups (top 20 by cumulative time):

Wed Sep 16 01:12:27 2026 output/fine/pbmc3k-rank_genes_groups.prof

42168 function calls (41011 primitive calls) in 1.205 seconds

Ordered by: cumulative time
List reduced from 1035 to 20 due to restriction <20>

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	1.205	1.205	__init__.py:83(fn_compatible)
1	0.000	0.000	1.205	1.205	_rank_genes_groups.py:469(rank_genes_groups)
1	0.001	0.001	1.168	1.168	_rank_genes_groups.py:398(compute_statistics)
5	0.003	0.001	1.139	0.228	_rank_genes_groups.py:277(wilcoxon)
5	0.006	0.001	0.977	0.195	_rank_genes_groups.py:45(_ranks)
4	0.000	0.000	0.918	0.229	generic.py:10012(rank)
4	0.000	0.000	0.918	0.229	generic.py:10127(ranker)
4	0.917	0.229	0.917	0.229	algorithms.py:1067(rank)
1	0.000	0.000	0.068	0.068	_rank_genes_groups.py:174(_basic_stats)
12	0.000	0.000	0.068	0.006	_index.py:29(_getitem_)
8	0.000	0.000	0.064	0.008	_csc.py:140(_get_arrayXslice)
8	0.000	0.000	0.064	0.008	_compressed.py:651(_minor_index_fancy)
60/59	0.000	0.000	0.060	0.001	indexing.py:1177(_getitem_)
41	0.000	0.000	0.058	0.001	indexing.py:1690(_getitem_tuple)
57	0.000	0.000	0.057	0.001	indexing.py:1720(_getitem_axis)
23	0.000	0.000	0.056	0.002	indexing.py:1206(_getbool_axis)
23	0.000	0.000	0.055	0.002	generic.py:4164(_take_with_is_copy)
8	0.055	0.007	0.055	0.007	{built-in method scipy.sparse._sparsetools.csr_column_index2}
23	0.000	0.000	0.055	0.002	generic.py:4049(take)
16	0.000	0.000	0.054	0.003	indexing.py:1005(_getitem_tuple_same_dim)

Console output (scanpy log):

```
(dependency deprecation warnings omitted)

... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 19024 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:00)
computing neighbors
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/_init_.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of obsm_keys. (e.g. `k` in adata.obsm` or `adata.obsm.keys() | {'u'} ``)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished: added to `.uns['neighbors']`
    `.obsp['distances']`, distances for each pair of neighbors
    `.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
    'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
    'X_umap', UMAP coordinates (adata.obsm) (0:00:05)
ranking genes
  finished: added to `.uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
```

pbmc6k: fine-grain profiling output

Dataset: pbmc6k

cProfile of sc.tl.rank_genes_groups (top 20 by cumulative time):

Wed Sep 16 01:12:48 2026 output/fine/pbmc6k-rank_genes_groups.prof

80033 function calls (78244 primitive calls) in 3.371 seconds

Ordered by: cumulative time

List reduced from 1035 to 20 due to restriction <20>

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	3.371	3.371	__init__.py:83(fn_compatible)
1	0.000	0.000	3.371	3.371	_rank_genes_groups.py:469(rank_genes_groups)
1	0.004	0.004	3.248	3.248	_rank_genes_groups.py:398(compute_statistics)
8	0.004	0.000	3.188	0.398	_rank_genes_groups.py:277(wilcoxon)
9	0.012	0.001	2.536	0.282	_rank_genes_groups.py:45(_ranks)
8	0.000	0.000	2.438	0.305	generic.py:10012(rank)
8	0.000	0.000	2.438	0.305	generic.py:10127(ranker)
8	2.436	0.304	2.436	0.305	algorithms.py:1067(rank)
155/154	0.001	0.000	0.263	0.002	indexing.py:1177(__getitem__)
1	0.001	0.001	0.257	0.257	_rank_genes_groups.py:174(_basic_stats)
96	0.001	0.000	0.255	0.003	indexing.py:1690(__getitem_tuple)
152	0.001	0.000	0.254	0.002	indexing.py:1720(__getitem_axis)
22	0.000	0.000	0.254	0.012	_index.py:29(__getitem__)
63	0.000	0.000	0.249	0.004	indexing.py:1206(_getbool_axis)
56	0.001	0.000	0.246	0.004	indexing.py:1005(__getitem_tuple_same_dim)
63	0.000	0.000	0.246	0.004	generic.py:4164(_take_with_is_copy)
63	0.001	0.000	0.244	0.004	generic.py:4049(take)
14	0.000	0.000	0.243	0.017	_csc.py:140(_get_arrayxslice)
14	0.001	0.000	0.243	0.017	_compressed.py:651(_minor_index_fancy)
63	0.000	0.000	0.242	0.004	managers.py:888(take)

Console output (scanpy log):

(dependency deprecation warnings omitted)

```
... reading from cache file cache/data-pbmc6k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:00)
computing neighbors
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/_init__.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of obsm_keys. (e.g. `k in adata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished: added to `uns['neighbors']`
    `.obsp['distances']`, distances for each pair of neighbors
    `.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:16)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:03)
```

pbmc10k: fine-grain profiling output

Dataset: pbmc10k

cProfile of sc.tl.rank_genes_groups (top 20 by cumulative time):

Wed Sep 16 01:13:27 2026 output/fine/pbmc10k-rank_genes_groups.prof

304961 function calls (299459 primitive calls) in 13.817 seconds

Ordered by: cumulative time
List reduced from 1064 to 20 due to restriction <20>

ncalls	totttime	percall	cumtime	percall	filename:lineno(function)
2/1	0.001	0.000	13.817	13.817	__init__.py:83(fn_compatible)
1	0.001	0.001	13.251	13.251	_rank_genes_groups.py:469(rank_genes_groups)
1	0.013	0.013	13.214	13.214	_rank_genes_groups.py:398(compute_statistics)
25	0.420	0.017	8.763	0.351	_rank_genes_groups.py:45(_ranks)
16	0.017	0.001	8.616	0.539	_rank_genes_groups.py:277(wilcoxon)
24	0.000	0.000	7.895	0.329	generic.py:10012(rank)
24	0.001	0.000	7.895	0.329	generic.py:10127(ranker)
24	7.888	0.329	7.889	0.329	algorithms.py:1067(rank)
1	0.014	0.014	2.702	2.702	_rank_genes_groups.py:174(_basic_stats)
54	0.001	0.000	2.672	0.049	_index.py:29(_getitem_)
30	0.000	0.000	2.607	0.087	_csc.py:140(_get_arrayXslice)
30	0.001	0.000	2.606	0.087	_compressed.py:651(_minor_index_fancy)
30	2.162	0.072	2.162	0.072	{built-in method scipy.sparse._sparsetools.csr_column_index2}
803/802	0.005	0.000	1.147	0.001	indexing.py:1177(_getitem_)
440	0.004	0.000	1.110	0.003	indexing.py:1690(_getitem_tuple)
800	0.004	0.000	1.108	0.001	indexing.py:1720(_getitem_axis)
360	0.003	0.000	1.079	0.003	indexing.py:1005(_getitem_tuple_same_dim)
367	0.001	0.000	1.076	0.003	indexing.py:1206(_getbool_axis)
367	0.002	0.000	1.060	0.003	generic.py:4164(_take_with_is_copy)
367	0.004	0.000	1.052	0.003	generic.py:4049(take)

Console output (scanpy log):

(dependency deprecation warnings omitted)

```
... reading from cache file cache/data-pbmc10k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.py:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:01)
computing neighbors
/home/jlim249/miniconda3/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/_init__.py:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of obsm_keys. (e.g. `k in adata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished: added to `uns['neighbors']`
    `.obsp['distances']`, distances for each pair of neighbors
    `.obsp['connectivities']`, weighted adjacency matrix (0:00:29)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 15 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:14)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:13)
```

Batch script: slurm_pbmc10k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-coarse
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-coarse-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
#SBATCH -c 12
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k
--out-dir ./output/coarse --num-threads 12
```

Batch script: slurm_pbmc10k_fine.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-fine
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-fine-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-fine-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
#SBATCH -c 12
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir ./output/fine
--num-threads 12 --profile-output ./output/fine/pbmc10k-rank_genes_groups.prof
```

Batch script: slurm_pbmc10k_instrumented.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-instr
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-instr-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-instr-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
#SBATCH -c 12
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="$MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/instrumented
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir
./output/instrumented --num-threads 12
```

Batch script: slurm_pbmc3k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-3k-coarse
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc3k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc3k-coarse-%A.err
#SBATCH --mem=16G
#SBATCH -n 1
#SBATCH -c 4
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k
--out-dir ./output/coarse --num-threads 4
```

Batch script: slurm_pbmc3k_fine.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-3k-fine
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc3k-fine-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc3k-fine-%A.err
#SBATCH --mem=16G
#SBATCH -n 1
#SBATCH -c 4
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir ./output/fine
--num-threads 4 --profile-output ./output/fine/pbmc3k-rank_genes_groups.prof
```

Batch script: slurm_pbmc3k_instrumented.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-3k-instr
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc3k-instr-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc3k-instr-%A.err
#SBATCH --mem=16G
#SBATCH -n 1
#SBATCH -c 4
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/instrumented
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir
./output/instrumented --num-threads 4
```

Batch script: slurm_pbmc6k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-6k-coarse
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc6k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc6k-coarse-%A.err
#SBATCH --mem=24G
#SBATCH -n 1
#SBATCH -c 8
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k
--out-dir ./output/coarse --num-threads 8
```

Batch script: slurm_pbmc6k_fine.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-6k-fine
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc6k-fine-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc6k-fine-%A.err
#SBATCH --mem=24G
#SBATCH -n 1
#SBATCH -c 8
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir ./output/fine
--num-threads 8 --profile-output ./output/fine/pbmc6k-rank_genes_groups.prof
```

Batch script: slurm_pbmc6k_instrumented.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-6k-instr
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc6k-instr-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc6k-instr-%A.err
#SBATCH --mem=24G
#SBATCH -n 1
#SBATCH -c 8
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/instrumented
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir
./output/instrumented --num-threads 8
```

pbmc3k: coarse profiling and instrumented timings

Dataset: pbmc3k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir  
./output/coarse --num-threads 4"
```

User time (seconds): 18.76

System time (seconds): 1.60

Percent of CPU this job got: 104%

Elapsed (wall clock) time (h:mm:ss or m:ss): 0:19.52

Average shared text size (kbytes): 0

Average unshared data size (kbytes): 0

Average stack size (kbytes): 0

Average total size (kbytes): 0

Maximum resident set size (kbytes): 888488

Average resident set size (kbytes): 0

Major (requiring I/O) page faults: 2775

Minor (reclaiming a frame) page faults: 153289

Voluntary context switches: 15803

Involuntary context switches: 372

Swaps: 0

File system inputs: 588064

File system outputs: 86512

Socket messages sent: 0

Socket messages received: 0

Signals delivered: 0

Page size (bytes): 4096

Exit status: 0

Instrumented timings:

section 1 elapsed: 1.590 seconds

section 2 elapsed: 4.912 seconds

section 3 elapsed: 0.001 seconds

section 4 elapsed: 0.102 seconds

section 5 elapsed: 0.047 seconds

section 6 elapsed: 0.020 seconds

section 7 elapsed: 0.030 seconds

section 8 elapsed: 0.252 seconds

section 9 elapsed: 3.348 seconds

section 10 elapsed: 0.088 seconds

section 11 elapsed: 5.828 seconds

section 12 elapsed: 1.307 seconds

pbmc6k: coarse profiling and instrumented timings

Dataset: pbmc6k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir  
./output/coarse --num-threads 8"
```

User time (seconds): 34.84

System time (seconds): 2.49

Percent of CPU this job got: 140%

Elapsed (wall clock) time (h:mm:ss or m:ss): 0:26.62

Average shared text size (kbytes): 0

Average unshared data size (kbytes): 0

Average stack size (kbytes): 0

Average total size (kbytes): 0

Maximum resident set size (kbytes): 984928

Average resident set size (kbytes): 0

Major (requiring I/O) page faults: 2805

Minor (reclaiming a frame) page faults: 215875

Voluntary context switches: 19308

Involuntary context switches: 237243

Swaps: 0

File system inputs: 672344

File system outputs: 197560

Socket messages sent: 0

Socket messages received: 0

Signals delivered: 0

Page size (bytes): 4096

Exit status: 0

Instrumented timings:

section 1 elapsed: 2.253 seconds

section 2 elapsed: 6.402 seconds

section 3 elapsed: 0.001 seconds

section 4 elapsed: 0.184 seconds

section 5 elapsed: 0.153 seconds

section 6 elapsed: 0.090 seconds

section 7 elapsed: 0.079 seconds

section 8 elapsed: 1.073 seconds

section 9 elapsed: 3.878 seconds

section 10 elapsed: 0.378 seconds

section 11 elapsed: 16.326 seconds

section 12 elapsed: 3.491 seconds

pbmc10k: coarse profiling and instrumented timings

Dataset: pbmc10k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir  
./output/coarse --num-threads 12"
```

User time (seconds): 63.20

System time (seconds): 6.63

Percent of CPU this job got: 124%

Elapsed (wall clock) time (h:mm:ss or m:ss): 0:56.27

Average shared text size (kbytes): 0

Average unshared data size (kbytes): 0

Average stack size (kbytes): 0

Average total size (kbytes): 0

Maximum resident set size (kbytes): 1890192

Average resident set size (kbytes): 0

Major (requiring I/O) page faults: 40

Minor (reclaiming a frame) page faults: 953916

Voluntary context switches: 24196

Involuntary context switches: 409586

Swaps: 0

File system inputs: 628960

File system outputs: 831456

Socket messages sent: 0

Socket messages received: 0

Signals delivered: 0

Page size (bytes): 4096

Exit status: 0

Instrumented timings:

section 1 elapsed: 2.275 seconds

section 2 elapsed: 6.459 seconds

section 3 elapsed: 0.001 seconds

section 4 elapsed: 0.757 seconds

section 5 elapsed: 0.873 seconds

section 6 elapsed: 0.448 seconds

section 7 elapsed: 0.176 seconds

section 8 elapsed: 1.308 seconds

section 9 elapsed: 30.960 seconds

section 10 elapsed: 0.803 seconds

section 11 elapsed: 14.895 seconds

section 12 elapsed: 13.733 seconds